

Public vessel-tracking data does not predict the Henry Hub–TTF spread: a pre-registered, power-bounded null

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Abstract

Positions of liquefied natural gas carriers are widely believed to carry information about the price gap between North American and European gas. A cargo that leaves the US Gulf today lands in Europe in two to three weeks, and that supply does not yet appear in any published statistic. We test this belief on the transatlantic trade. From 54,494,686 AIS position fixes (2016-01-01 to 2026-08-10) we reconstruct 87,874 port events at 40 terminals and a 34-signal daily panel, built once with hindsight for validation and once strictly point-in-time for modelling. The US side of the reconstruction is checked against EIA export totals. We then ask whether tanker-derived signals predict the one- and four-week change in the Henry Hub–TTF spread after controlling for its own persistence, weather, storage and oil. Each hypothesis, sign and acceptance bar was recorded in a dated append-only log before the corresponding fit. We find no predictive content. A controls-only model performs worse than a no-change forecast (−9.3% at one week, −18.2% at four). No signal reaches a Newey–West $|t|$ of 1.96 (maximum 1.50). Three pre-registered mechanisms that a linear scan could miss (the Europe-bound share of cargo, an interaction with EU storage, and a tightness-regime split) fail under a Bonferroni bar with an untouched holdout. The design can detect effects of at least 1.72% of residual variance; the largest observed is 0.009. The result therefore excludes effects large enough to trade and says nothing about smaller ones. Three things stand independently of the null: a decade-scale event reconstruction with a measured capture rate, two construction hazards specific to voyage-derived features (the unit of observation is defined only in retrospect, and feature availability changes across data vintages, so that a naive loader leaked 65% of the sample), and a physical-nowcast benchmark showing that a naive moving average is near-optimal for weekly US loadings.

1 Introduction

The Henry Hub–TTF spread is the price of moving a unit of gas from the United States to Europe, and liquefied natural gas (LNG) carriers are the means by which it moves. A laden carrier that leaves Sabine Pass arrives at Rotterdam roughly fourteen to eighteen days later. Between departure and arrival its cargo is committed, is visible to anyone with an AIS receiver, and is absent from every published supply statistic. It is therefore natural to expect vessel positions to lead the spread. This expectation is widely held and is the basis of a commercial data industry.

This paper tests it and finds no support. An efficient-markets view predicts that outcome, so the null on its own would be a thin result. The paper makes four contributions, of which the null is one.

A validated decade-scale reconstruction. Section 2 describes a pipeline that turns 54,494,686 raw position fixes and two public archives into 87,874 port events at every LNG terminal in seven US and European zones, using hand-drawn berth, anchorage and approach polygons and a per-vessel state machine. The US side is checked against EIA monthly export totals and the check is reported as a capture rate by year. A live acquisition system, which allocates a scarce budget of AIS subscription slots across the world fleet and buys back positions for vessels that fall silent near a terminal,

is described because most of the engineering effort went there and because its coverage limits are part of the result.

Two construction hazards specific to voyage-derived features. Section 3 shows that the natural unit of observation, a voyage, does not exist until both of its endpoints have been observed, so a statistic computed over completed voyages embeds the future unless the population is explicitly bounded. At a 2020 as-of date a straightforward loader returned 20,351 legs, of which 13,130 (65%) had been closed by arrivals that had not yet occurred. Separately, the availability of a feature can change across data vintages. The same loader attached 2,101 destination declarations to 2020 voyages from a data feed that began in 2026. A date filter on observations cannot catch this, because the individual rows are correctly dated; the table itself should not exist at that vintage.

A pre-registered, power-bounded null. Three features of the design distinguish the result from a mere absence of findings. First, every target, control set, lag, sign and acceptance bar was written into a dated, append-only decision log before the corresponding fit, so no choice reported below was made after seeing its outcome. Where a pre-registered sign was falsified, it is reported as falsified. Second, the inference respects the structure of the data. Weekly changes over a four-

week horizon overlap, so all standard errors are Newey–West (a covariance estimator that corrects for serial correlation in the residuals; Section 6) with the lag fixed in advance at $h + 1$ weeks. Training rows whose target reaches into the test week are purged. The three follow-up mechanism tests are held to a Bonferroni bar (the significance threshold divided by the number of tests) and must replicate their sign on a holdout that was not examined until they had passed on the discovery sample. Third, the study bounds what it could have found. With 327 weekly observations the design detects effects of at least 1.72% of residual variance, and the largest observed anywhere is 0.009. The result excludes effects large enough to trade and is silent on smaller ones.

Usable by-products. The physical nowcasts of Section 5 establish that a four-week moving average is close to optimal for weekly US loadings, and say why. The outage rule of the same section is a monitor with a stated recall and delay. The vintage log of Section 3 is a cheap check that, to our knowledge, is rarely kept.

The paper proceeds as follows. Section 2 describes the pipeline, the live acquisition system and the ground-truth check. Section 3 explains how the signal panel is built point-in-time. Section 4 describes the pre-registration protocol and its limits. Section 5 reports the physical nowcasts. Section 6 reports the spread tests. Section 7 bounds their power. Section 8 interprets the results and lists what the work found besides the null, Section 9 sets out what would change the answer, and Section 10 qualifies everything. Appendix C defines the statistical terms used.

2 Data and pipeline

2.1 Sources

Vessel positions come from three feeds of different fidelity, and no signal is aggregated across the boundary between them. For the United States we use the NOAA Marine Cadastre archive of terrestrial Class-A AIS [10], which is exhaustive along the US coast from 2016 and ends on 2025-12-31. For Europe and the open ocean we use Global Fishing Watch port-visit and voyage events [5]. These supply arrivals and departures but carry no coordinates and no anchorage events; the feed ends on 2026-02-21. A live subscription to an AIS aggregator covered a tier-ranked fleet from April 2026 (Section 2.3) and is used only for the vintage self-validation described in Section 3.

The fleet in scope is every LNG carrier and floating storage and regasification unit (FSRU) in the IGU fleet list, 828 hulls, resolved to an MMSI through a vessel-database lookup. The signals concern the transatlantic subset of their movements: laden departures from US export terminals, arrivals and discharges at European import terminals, and the ballast return. Cargoes bound for Asia or Latin America enter the panel only as an unresolved-destination band (Section 6).

Prices and controls are public. Henry Hub spot and Lower-48 storage are from the EIA API [12]. The TTF front-month contract is from a keyless quote feed and was cross-checked against the World Bank’s monthly European gas series [13]; the regression slope is 0.999 with $R^2 = 0.9999$ over 106 months, so the distortion introduced by the continuous roll is negligible. EUR/USD and Brent are from FRED [2]. EU storage is from GIE AGSI+ [4]. Heating and cooling degree days are computed from ERA5 reanalysis temperatures at weighted demand centres [11].

2.2 Reconstruction

Table 1 gives the scale of the reconstruction and Figure 1 its shape. Fixes are resolved against 142 polygons at 40 terminals in seven zones (US Gulf, US Atlantic, north-west Europe, Baltic, Iberia, western Mediterranean and eastern Mediterranean). The polygons were drawn by hand in QGIS from satellite imagery and port charts, one berth, one anchorage and one approach envelope per terminal, with sub-zones where a terminal has several berths. The approach envelope is deliberately drawn to contain the anchorage, the channel and the berth, so that a single visit stays open while a vessel transits between them; without it the four-to-six hour channel transit at Sabine Pass or Rotterdam would register as a departure and a fresh arrival. Figure 2 shows the three polygons at Sabine Pass with a sample of the events emitted inside them.

A per-vessel state machine (Figure 3) walks each vessel’s fixes in time order and emits zone entry, anchorage, mooring and departure events. A vessel is *anchored* after at least 30 minutes inside an anchorage polygon at speed over ground below 1 kn, *moored* after the same dwell inside a berth polygon, and *departed* after 15 minutes outside the berth at more than 1 kn. Anchorage entries and exits are raw polygon crossings with no dwell filter, so that queue times can be measured without the half-hour back-dating of the dwell-confirmed marker. A vessel whose first observed fix is already inside a polygon receives synthetic entry and mooring events flagged as a cold start.

Laden state is inferred from the draught the vessel broadcasts relative to its design draught,

$$\text{laden}_e = \mathbf{1}[d(t_e) \geq 0.85 d_{\text{design}}], \quad (1)$$

where $d(t_e)$ is the most recent reported draught at or before the event time for inbound events and, for departures, the first draught reported in a window from one hour before to six hours after the event, because a master typically rebroadcasts the lighter draught only after undocking. When no usable draught exists the state is set from the terminal’s flow direction: a vessel that moored at an export terminal leaves laden, and one that moored at an import terminal leaves in ballast.

Three pairings of those events underlie everything that follows and are drawn in Section 3. A *leg* pairs a departure with the vessel’s next zone entry and represents a voyage. A *visit* pairs a mooring with the next departure and represents berth occupancy. A *queue*

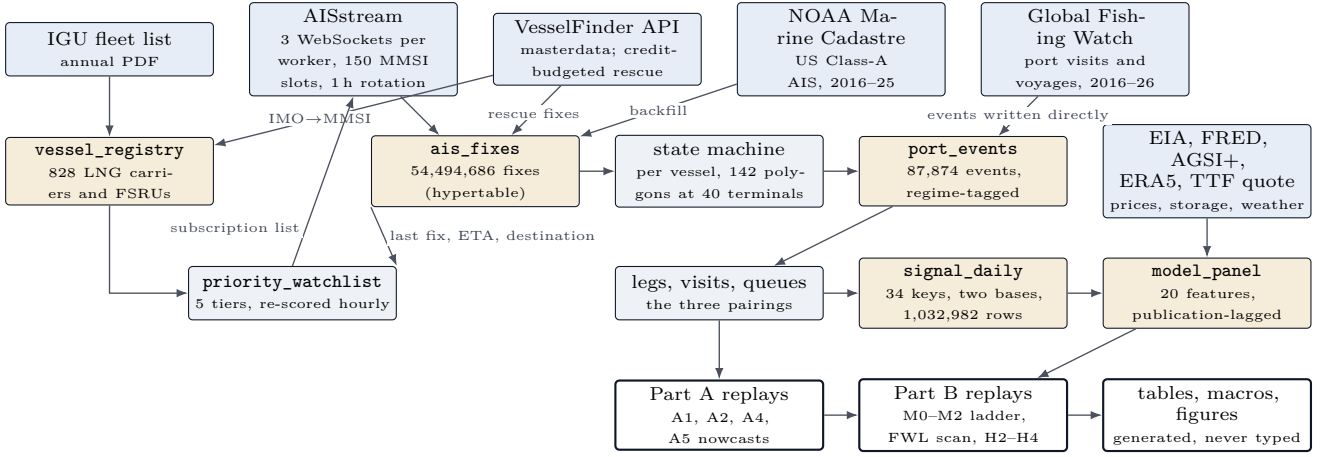


Figure 1: The pipeline from sources to paper. Blue boxes are external sources, gold boxes are stored tables, grey boxes are derivations, and the outlined boxes are the replays that generate every table and figure. The watchlist closes a loop: the ranking computed from stored fixes decides which vessels the live feed subscribes to next.

Table 1: Reconstruction inventory. Port events are labelled by the feed that produced them. The two backfill feeds end months before the live feed, which determines the sample choice in Section 6.

layer	rows	span
AIS position fixes	54,494,686	2016-01-01 to 2026-08-10
port events (state machine)	87,874	
noaa	44,363	to 2025-12-31
gfw	40,273	to 2026-02-21
mmsi_filter	1,974	to 2026-08-10
bbox	1,264	to 2026-05-30
in-scope fleet (LNG carriers + FSRUs)	828	
laden US export departures	9,855	
signal panel rows (signal_daily)	1,032,982	34 keys, two bases
model panel rows (model_panel)	77,500	20 features

pairs the first anchorage entry with the next mooring at the same terminal and represents the wait to berth.

2.3 Live acquisition

The live feed is the part of the project that took most of the engineering, and its limits are part of the result, so it is described here even though the decade panel is what the tests run on. The AIS aggregator caps a subscription at 50 vessel identifiers per connection. The system therefore runs two ingester workers on separate egress addresses, each holding three WebSocket connections: two connections form a persistent block of 100 slots and the third is a scan rotation of 50 slots that is re-drawn every hour, for 300 slots in total against a fleet of 828.

Which vessels hold a slot is decided by a ranking recomputed hourly from stored fixes and static data (Figure 4). Tier 1 is any vessel seen inside a terminal polygon in the last three days. Tier 2 is a vessel with a parsed ETA within 48 hours or a declared destination that resolves to a known terminal. Tier 3 is a vessel inside the wider zone bounding box, ordered by whether it is closing on a terminal. Tiers 4 and 5 are recently and not recently seen. Two classes are pinned into the persistent block regardless of tier: a vessel on an open leg inside its expected arrival window in either direction, and a vessel currently in port, so that a long berth queue does not decay out of coverage. FSRUs are demoted to the lowest tier because a deployed FSRU sits moored

for months and its own fixes never drive a signal. At the time of writing 261 of 827 ranked vessels hold a slot: 34 pinned, 127 persistent and 100 in rotation.

Terrestrial AIS does not see the open ocean, and a vessel that leaves a slot goes dark until it is re-ranked. Two backstops exist. First, a vessel that goes silent near a terminal while a leg-defining event is at risk (a laden departure, an import arrival, a berthing) is looked up on a commercial vessel-position API and the returned position is injected as an ordinary fix, so the state machine re-acquires it for free. The lookups cost credits from a reserve that expires on a fixed date, so the daily budget is derived each run as the straight line from the current balance to zero at expiry; leg-defining classes are exempt from the line and lower classes spend only the surplus above it (Figure 5). Between the first snapshot and 2026-08-10 the reserve fell from 4,947 to 3,801 credits against an expiry of 2027-06-02, over 2,072 lookups of which 1,041 returned a usable position; 2,133 candidate lookups were declined by the budget, which is the measured unmet demand. Second, a daily discovery pass resolves newly delivered hulls from the fleet list, because under a closed subscription list no vessel can otherwise enter the system.

Two limits follow directly. A vessel that was never ranked highly enough to hold a slot leaves no live trace at all, and a vessel that drops out of coverage mid-voyage is only recovered if it re-enters a slot or is bought back. Both are absent from the decade archives, which is why

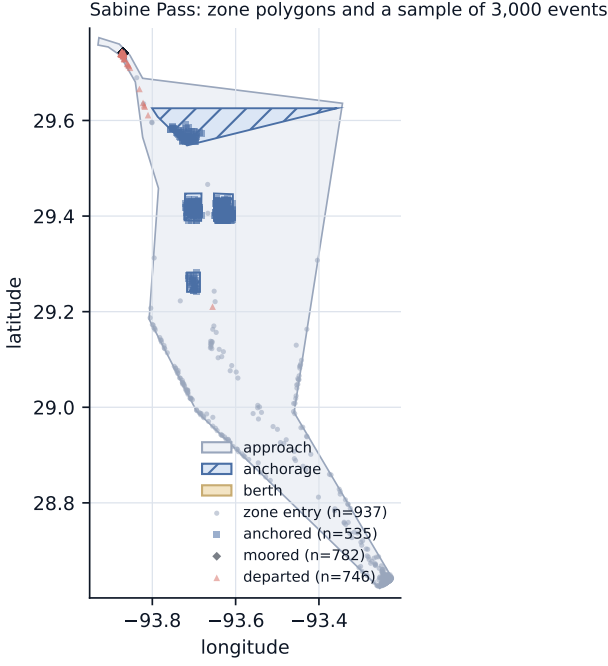


Figure 2: The hand-drawn polygons at Sabine Pass and a random sample of the 18,878 coordinate-bearing events the state machine emitted there. The approach envelope (light) contains the anchorage (hatched) and the berth (gold, at the head of the channel). Zone entries scatter along the envelope boundary, anchored events cluster in the anchorage boxes, and moored and departed events sit on the berth.

the tests of Sections 5 and 6 run on the NOAA and GFW backfills and the live feed serves only to validate the point-in-time construction.

2.4 Ground truth

The US side of the reconstruction can be checked against a published total. The EIA reports monthly US LNG exports X_m in MMcf of gas. Dividing by a nominal cargo of $V_c = 174,000 \text{ m}^3$ of LNG, which at a volumetric expansion of 600 is about 3.69 Bcf of gas, gives an implied cargo count, and the ratio of observed laden departures D_y to that count is the capture rate for year y :

$$c_y = \frac{D_y}{\sum_{m \in y} X_m / (600 V_c)}. \quad (2)$$

Restricted to the NOAA feed, c_y is 46% in 2020 and rises to 93%, 103% and 105% in 2023, 2024 and 2025 (Table 2, Figure 6). The gradient is a property of the archive, not of the vessels: the NOAA archive is assembled from terrestrial receivers whose density along the Gulf coast grew over the decade, and a cargo that loaded in 2020 out of range of a receiver left no fix. The GFW feed has the opposite gradient, strong in the years the NOAA archive is thin and fading after 2022, because it is fed by satellite receptions whose coverage of the fleet declined. The two feeds are not fused in this paper. The project’s diagnostic tool counts every feed and therefore double-counts departures seen by both, and a deduped union calibrated to the EIA total is listed as future work (Section 9).

The consequence for modelling is stated here because

Table 2: NOAA-only capture of US LNG export cargoes relative to the EIA-implied count, Equation (2).

year	captured	EIA-implied	capture
2016	48	51	95%
2017	159	192	83%
2018	216	294	74%
2019	290	494	59%
2020	300	648	46%
2021	715	966	74%
2022	918	1048	88%
2023	1,099	1178	93%
2024	1,216	1185	103%
2025	1,570	1494	105%

it matters everywhere below. Early-year levels of any US signal are biased low by up to half. The models therefore never treat an early-year level as a physical quantity. The physical nowcasts (Section 5) forecast within-window changes and score against nulls computed from the same observed series, so a uniform under-count deflates the model and the null identically. The spread scan (Section 6) residualises each signal on the controls and on its own past before it is used, so a slowly varying level bias is absorbed by the persistence term. What neither design can absorb is a level-dependent non-linearity in the early years, and that is listed as a limitation.

Figure 7 shows the target and the headline signal on one timeline. The two structural events that any model must survive are visible: the 2020 and 2021 inversions, in which Henry Hub briefly exceeded TTF, and the 2022 European crisis.

3 Building the panel point-in-time

Figure 8 shows the three pairings on one vessel’s year. Every signal is built from them twice. A *physical* basis uses hindsight and serves validation. A *knowable* basis uses, on each day d , only what could have been observed on day d , and serves modelling.

The headline signals are gas volumes. Let leg i carry capacity V_i (m^3 of LNG), depart at t_i^{dep} and arrive at t_i^{arr} . The physical at-sea stock on day d is

$$S_d^{\text{phys}} = \sum_i V_i \mathbf{1}[t_i^{\text{dep}} \leq d < t_i^{\text{arr}}], \quad (3)$$

banded by the arrival zone. The knowable stock replaces the arrival test with what is known on d : a leg is at sea if it has departed and no arrival has been *observed* by d , and it is banded by destination only if a destination was declared by d , otherwise it falls in the unresolved band. Loading and discharge rates are amortised flows: a visit that berths for H_j hours deposits V_j/H_j per hour across those hours, so that a closed visit integrates to exactly one cargo,

$$L_d = \sum_j V_j \frac{|[d, d+1) \cap [t_j^{\text{moor}}, t_j^{\text{dep}})|}{H_j}, \quad (4)$$

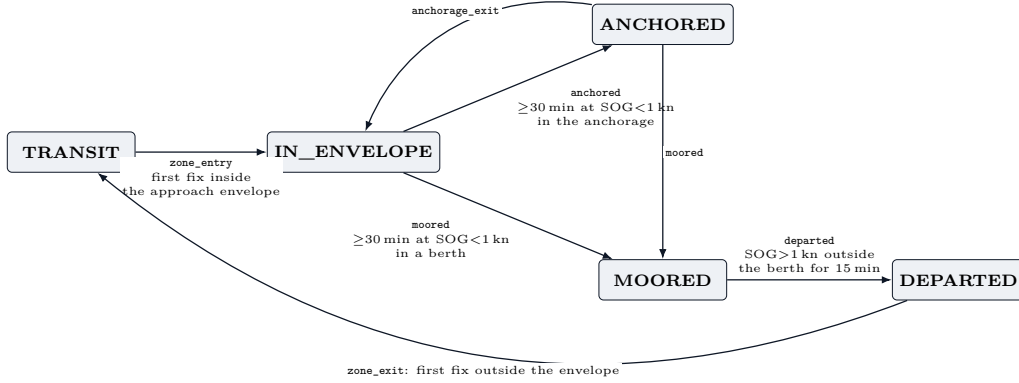


Figure 3: The per-vessel state machine. Each fix is first resolved to a polygon (berth wins over anchorage, which wins over approach; an open visit is sticky to its terminal), then the state is advanced. Dwell-confirmed events are back-dated to the first qualifying fix. Transitions require a fix outside the relevant polygon, so a gap in coverage never emits an event on its own.

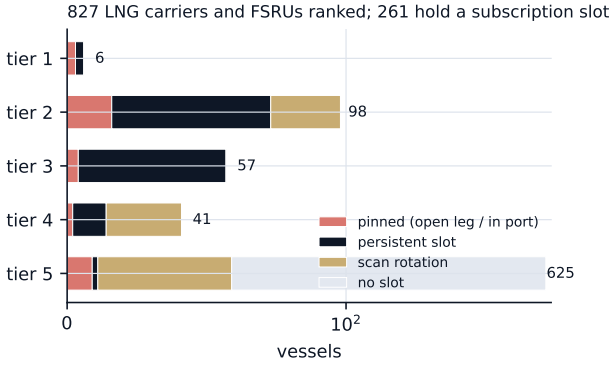


Figure 4: Subscription slots by tier at the time of writing. The horizontal axis is symmetric-log so that the 566 unslotted vessels in tier 5 do not hide the allocation in tiers 1 to 4.

and an open visit estimates H_j from the terminal’s closed-visit mean until it departs, capped at one cargo. Durations (queue waits, voyage ages) are averaged across terminals with weights equal to the number of legs or visits in each cell, $\bar{x}_d = \sum_k n_k x_k / \sum_k n_k$; volumes are summed.

Two properties of voyage-derived features make the knowable basis difficult to construct correctly. Both were discovered during the project rather than anticipated, and Figure 9 draws them.

3.1 The unit of observation is defined in retrospect

A voyage leg does not exist as a *completed* unit until both a departure and the next arrival have been observed. At time t , a vessel that departed five days earlier is an open leg of known age and unknown duration and destination; that open leg is itself a feature, and the at-sea stock of Equation (3) counts it from the day it departed. What does not yet exist is any statistic that conditions on completion: its duration, its arrival zone, its floating-storage label. In hindsight the same vessel has a complete leg with all three. Any statistic over legs therefore conditions on completion unless the pop-

ulation is explicitly bounded to what was observable at d ,

$$\mathcal{L}(d) = \{i : t_i^{\text{dep}} \leq d\}, \quad (5)$$

t_i^{arr} attached only if $t_i^{\text{arr}} \leq d$.

Figure 10 quantifies the effect. At an as-of date of 1 June 2020, a loader that pairs all departures with all arrivals and then filters by date returns 20,351 legs. Of these, 13,130 (65%) are closed by arrivals that had not yet occurred. With the population bounded by Equation (5), 4,713 legs remain, none of them closed by the future. Status labels move with the bound: the number of legs classified as floating storage fell from 95 to 9, because that label is assigned in hindsight. At an as-of date after the data maximum the two loaders agree on every count, which confirms that the bound is a restriction of the same query and not a different one.

3.2 Feature availability changes across vintages

The same naive loader attached 2,101 destination declarations to the 2020 legs. Declarations come from a live AIS static-data feed that began in 2026, so no declaration existed in 2020. This failure differs from the first. A date filter on the observations is satisfied, because each declaration row carries a timestamp within the vessel’s current voyage, but the table being joined should not exist at that vintage. The bound therefore reconstructs the declaration from the raw static messages available at d , and at a 2020 as-of date it returns none.

3.3 What the knowable basis bounds

The knowable basis bounds on event time. Event timestamps are back-dated to the fix at which a transition first qualified, so a departure is dated to the vessel’s last fix inside the berth polygon and not to the later fix that confirmed it had left. The gap between the two is the feature’s own publication lag. We measured it directly. The first fix after a departure event arrives at a median lag of zero hours, and more than 24 hours later for

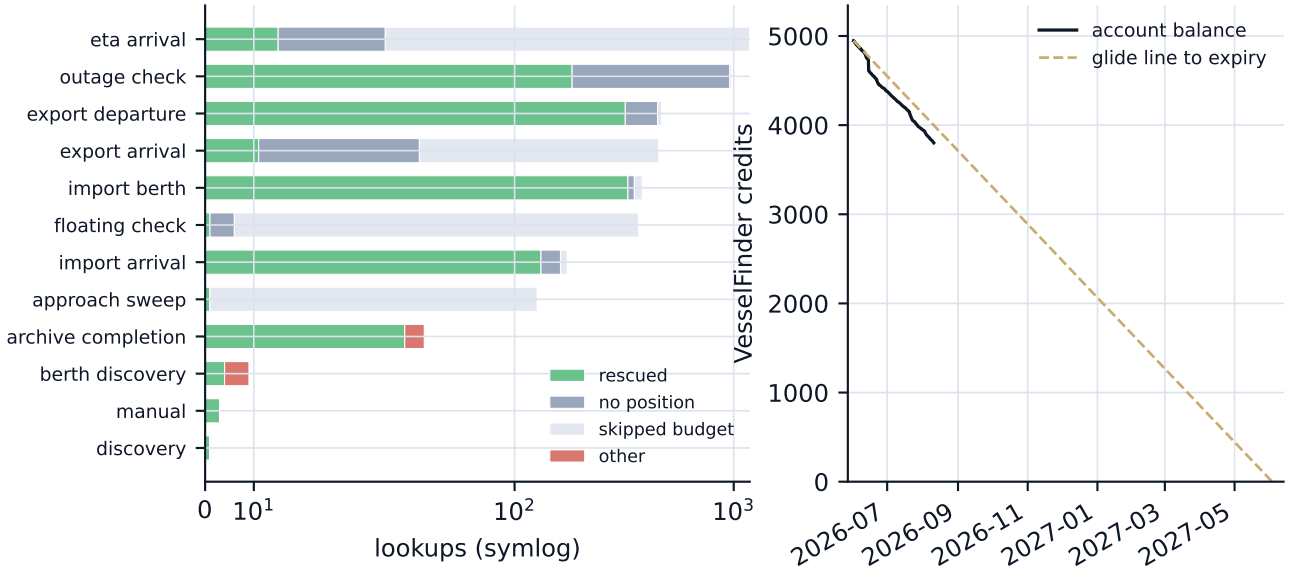


Figure 5: The credit-budgeted position backstop. Left: outcomes of every lookup attempt by rescue class, on a symmetric-log axis; green is a usable position, grey is a lookup that returned nothing, and light grey is a candidate the budget declined. Right: the account balance against the glide line from the first snapshot to zero at expiry, which is the policy the daily cap is derived from.

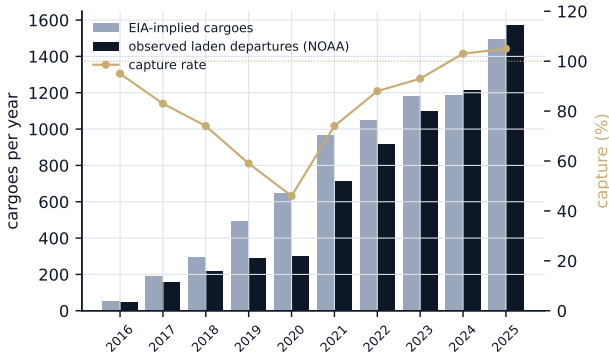


Figure 6: Observed laden departures against the EIA-implied cargo count by year. The capture rate (gold, right axis) is the coverage of the NOAA archive, which is why early-year levels of every US signal are biased low.

0.3% of departures on the NOAA feed ($n = 1,500$) and 1.5% on the live feed ($n = 343$). On the decade panel the lag is immaterial. On the live feed it is a genuine data limitation: a vessel that drops out of terrestrial coverage as it leaves the berth is only back-dated once it reappears, and a satellite feed would remove the gap. This is one of the reasons the live acquisition system of Section 2.3 spends most of its effort on keeping vessels near terminals in a slot.

A separate log records the knowable value of every signal as it was printed on each day of live operation, so that the bound can be checked rather than assumed. Comparing the rebuilt knowable series with those prints on days old enough to have settled, the check flags 68 rows whose value moved after a next-day print. All of them lie in the live tail and are consistent with late-arriving fixes back-dating departures. A further 909 rows differ from prints taken during the day, before that day's fixes had finished arriving. That comparison measures intraday timing and not leakage; it is reported but not scored. The live tail lies outside the modelling

sample used in Section 6.

3.4 The modelling grid

Prices, storage and the tanker signals arrive on different calendars. They are assembled onto one daily grid by forward-fill under a per-cadence staleness ceiling. An inner join is never used: Henry Hub spot prints on 69% of calendar days, and an inner join across the price legs would drop a third of the history without any indication that it had done so. Each external series is shifted to the date on which it became public (EIA weekly storage by six days, daily spot by one, AGSI+ by one). The tanker signals are shifted by zero, because the knowable basis is already point-in-time and a second shift would double-count. Degree days are not shifted. Their role is to be partialled out, so the relevant quantity is the weather that occurred and not the weather that was forecast. Figure 11 shows the eight signals that enter Section 6, on this grid.

3.5 The coverage seam

The two backfill feeds end on 2025-12-31 (NOAA) and 2026-02-21 (GFW). After those dates the signals are carried by the live feed alone, which contributes 3,238 events against 84,636 from the backfills (Figure 12). The EU-bound stock at sea falls from 4.80 to 0.41 million m³ between December 2025 and January 2026, a 91% drop with no physical cause. This was found by plotting Figure 7 and was not caught by any automated check. The primary modelling sample is therefore truncated at 2025-12-31. Full-panel results are reported in Appendix B and change no conclusion.



Figure 7: The Henry Hub–TTF spread (top) and EU-bound gas at sea (bottom), daily, on the coverage-consistent sample used for modelling. The extremes coincide with Winter Storm Uri (February 2021) and the August 2022 TTF peak. The collapse of the at-sea stock in the final weeks is not a physical event: it is the point at which the two backfill feeds end and only the thin live feed remains (Section 3.5), which is why the primary sample is truncated there.

4 Pre-registration protocol

The project keeps a dated, append-only decision log. A hypothesis, target, lag, sign or acceptance bar is recorded on the day its entry is written and before the corresponding fit. A later change is made by a new, dated entry that names the result which prompted it. The log is what allows a reader to distinguish a test that found nothing from a search that stopped at the first result to cross a threshold. Table 3 lists the entries that fixed each model reported below.

Two consequences of the protocol appear in the results. A pre-registered sign that the data falsified is reported as falsified (Section 5). A test that produced an apparent 87% improvement over its null was withdrawn, because the protocol’s own comparability condition failed (Section 5). Neither outcome would have been recorded without a criterion fixed in advance. A third consequence is smaller but real: an audit of the log before this paper was drafted found that one synthesis entry had swapped two models’ scores, and the rule that every number in the text is generated from the replay output surfaced the discrepancy the first time the table was read next to the prose.

Limits of the log. The log is a text file in the project’s version control, and its git history does not independently establish the ordering it asserts. The entries for the physical nowcasts (D-001 through D-026) were committed together, weeks after the work they describe. The two spread pre-registrations (D-028 and D-031) were each committed in the same change as their result. The content of each pre-registration, meaning

the signs, bars, horizon and holdout fixed in its text, was written before the corresponding run, but a sceptical reader has only the author’s word for this. We state it here rather than leave it to be found. A third-party timestamped registry would have cost nothing and will be used for any future entry, which will be committed and pushed before the run it governs.

5 Physical nowcasts

Before asking whether the signals predict a price, we ask whether they predict the physical quantities they measure directly: weekly EU arrivals, weekly US loadings and terminal outages, at one- and two-week horizons. These targets have high signal-to-noise ratios and published ground truth, and they establish what the pipeline can and cannot see before any price enters the analysis. The models are labelled A1 to A5 in the order they were pre-registered in the project plan (Part A of the plan is the physical layer, Part B the spread); A3, a survival model of voyage time, was specified but not built, for reasons recorded in the log. Four models were pre-registered and scored walk-forward against two naive nulls, and each was designed to address the diagnosed weakness of the previous one.

Targets, nulls and scoring. Time is a weekly grid of Mondays from 2018. For an as-of date t the forecast windows are $W_k = (t + 7(k - 1), t + 7k]$ days, and the target is a count of cargoes in the window: the conditional EU arrival count for A1 (laden legs already at sea at t whose first European zone entry falls in W_k)

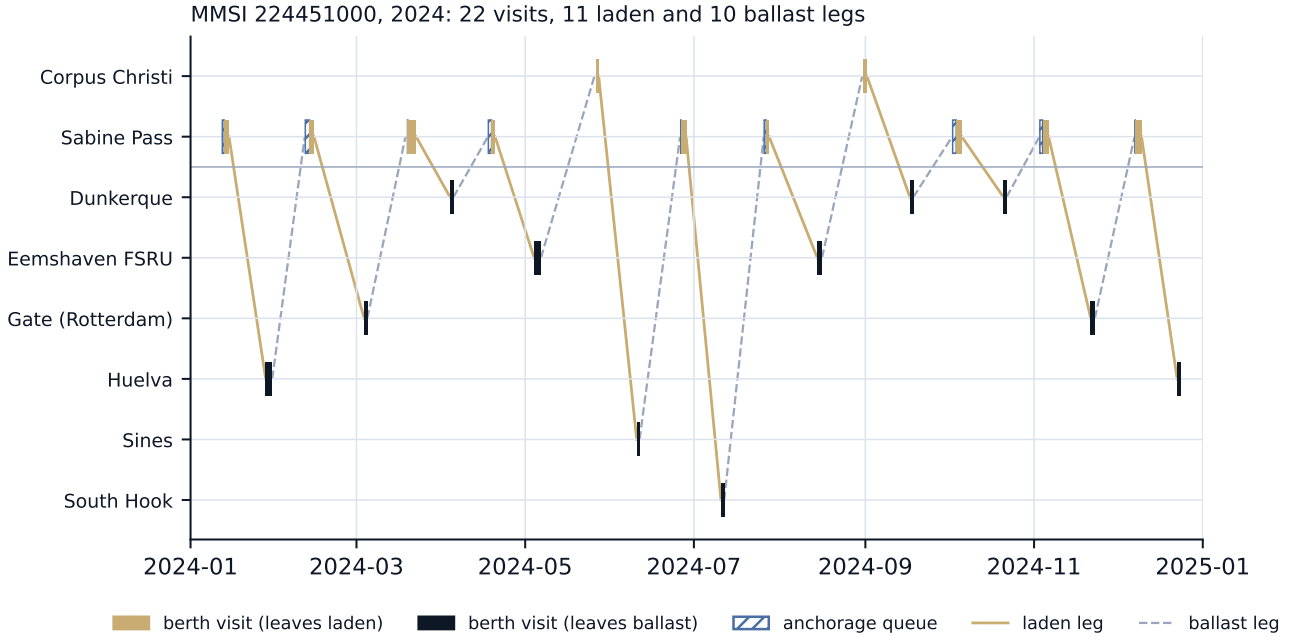


Figure 8: One carrier’s 2024 as the three pairings. Each row is a terminal, US export terminals above the rule and European import terminals below. A gold bar is a berth visit that ends with a laden departure, a black bar one that ends in ballast, and a hatched bar the anchorage wait before berthing. Solid lines are laden legs from a US departure to the next European zone entry; dashed lines are the ballast return. This vessel made 22 visits, 11 laden and 10 ballast legs, from 108 events; the US events are from the NOAA archive and the European ones from GFW, which is why the European visits carry no anchorage wait.

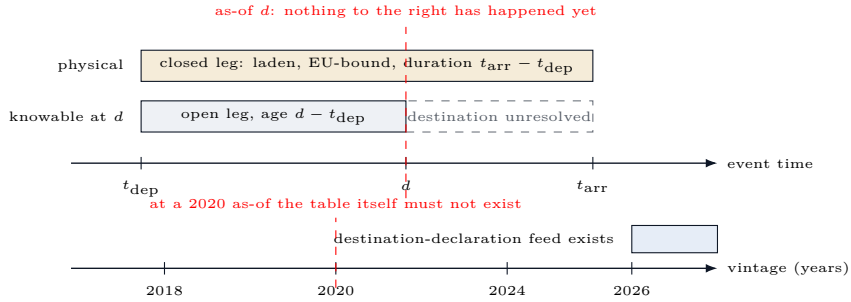


Figure 9: Top: one leg on the event-time axis. In hindsight it is a closed, EU-bound leg of known duration; at an as-of date d before arrival it is an open leg of known age and unknown destination, and the at-sea stock counts it as such. Bottom: the vintage axis. A table that begins to exist in 2026 must return nothing at a 2020 as-of date even though every row in it is correctly dated.

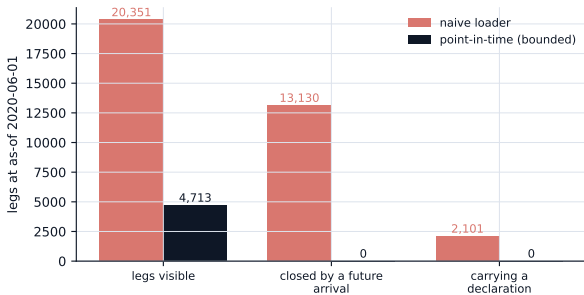


Figure 10: Voyage legs visible at an as-of date of 1 June 2020 under the naive loader and under the point-in-time bound. Recomputed from the event tables when the figure is generated.

and the US laden-departure count for A2 and A4. The *persistence* null is the last fully elapsed week’s count, and the *climatology* null is the trailing four-week mean; both are computable at t . Models are scored by mean absolute error (MAE) in cargoes, and a model *beats* a

null when its MAE is lower. We report each model’s excess error over the climatolgy null,

$$\text{excess} = \frac{\text{MAE}_{\text{model}} - \text{MAE}_{\text{clim}}}{\text{MAE}_{\text{clim}}}, \quad (6)$$

so that a positive value is worse than the null. Calibration is scored by the share of outcomes inside the model’s central 80% interval, which should lie between 70 and 90%. The pre-registered bar was to beat both nulls on one-week MAE and to land inside that coverage band. Figure 13 shows the loadings series the US-side models forecast, with both nulls drawn on it.

A1: duration climatolgy. For each open laden leg i of age a at t , the probability that it produces an observed European arrival in W_k is estimated directly as a count ratio over the matured population of the

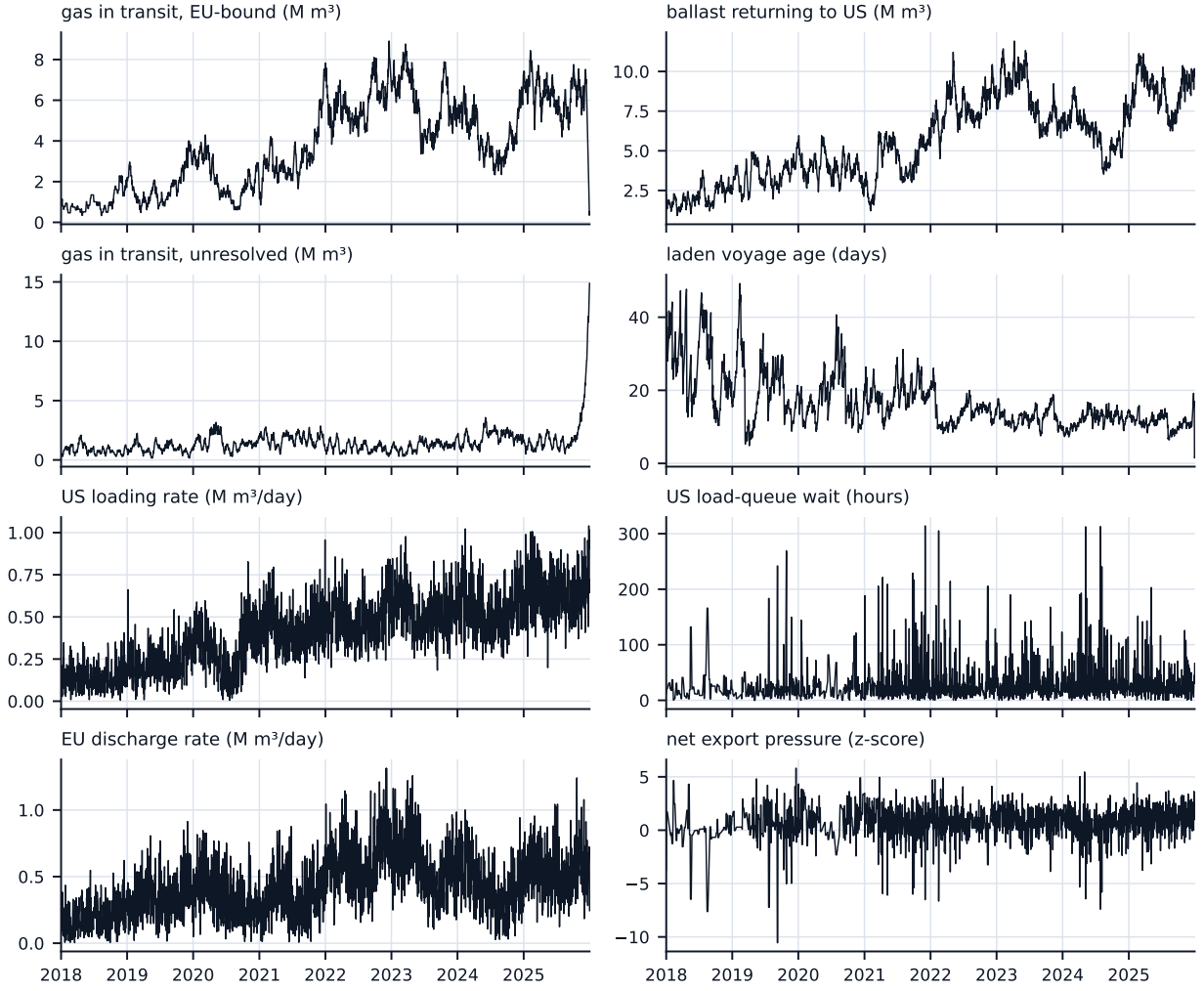


Figure 11: The eight pre-registered tanker signals, daily on the knowable basis, 2018 to the coverage horizon. The upward drift in the volume signals is the growth of US export capacity together with the capture gradient of Figure 6; the spread scan removes both through the persistence term. The step at the very end of the EU-bound and unresolved in-transit panels is the coverage seam.

trailing year,

$$p_i(W_k) = \frac{N_{\text{eu}}(\text{duration} \in (a + u_0, a + u_1])}{N(\text{open at age } a)}, \quad (7)$$

where $(u_0, u_1]$ is the window offset. The ratio factorises into an age-conditional probability that the leg is Europe-bound at all and the duration distribution of Europe-bound legs, but because both come from one population the factors telescope into the single ratio above. The weekly count is the Poisson-binomial sum over open legs, with mean $\sum_i p_i$ and an exact probability mass function by convolution. A1 has no fitted parameter.

A2: negative-binomial count model. Weekly US loadings y are modelled as

$$\log \lambda = \beta_0 + \sum_j \beta_j x_j, \quad y \sim \text{NB}(\lambda, k), \quad (8)$$

with five features fixed in advance together with their expected signs: last week’s count, the trailing four-week mean, the number of vessels in berth, ballast arrivals in the past week, and the anchorage queue depth. The

first two features are the two nulls, so the scorecard asks whether the three physical features add anything to the autoregression. Coefficients are refit every week on an expanding window with a minimum of 104 weeks.

A4: Kalman local level. The same target is modelled as a latent weekly rate observed with noise,

$$\begin{aligned} x_t &= x_{t-1} + w_t, & w_t &\sim N(0, q), \\ y_t &= x_t + v_t, & v_t &\sim N(0, r), \end{aligned} \quad (9)$$

with (q, r) by exact maximum likelihood, walk-forward. In steady state the filter’s one-step forecast is an exponentially weighted moving average, $\hat{x}_t = \alpha y_t + (1 - \alpha)\hat{x}_{t-1}$, with α set by the ratio q/r and an effective window of $(2 - \alpha)/\alpha$ weeks. A4 therefore asks whether an optimally tuned moving average beats the fixed four-week one.

Results. Table 4 and Figure 14 give the one-week results. A1 is +27.5% worse than the climatology null; its estimator is built from legs at least ninety days old inside a one-year window, so it is stale, and its errors are the classic lag pattern of under-shooting on the way

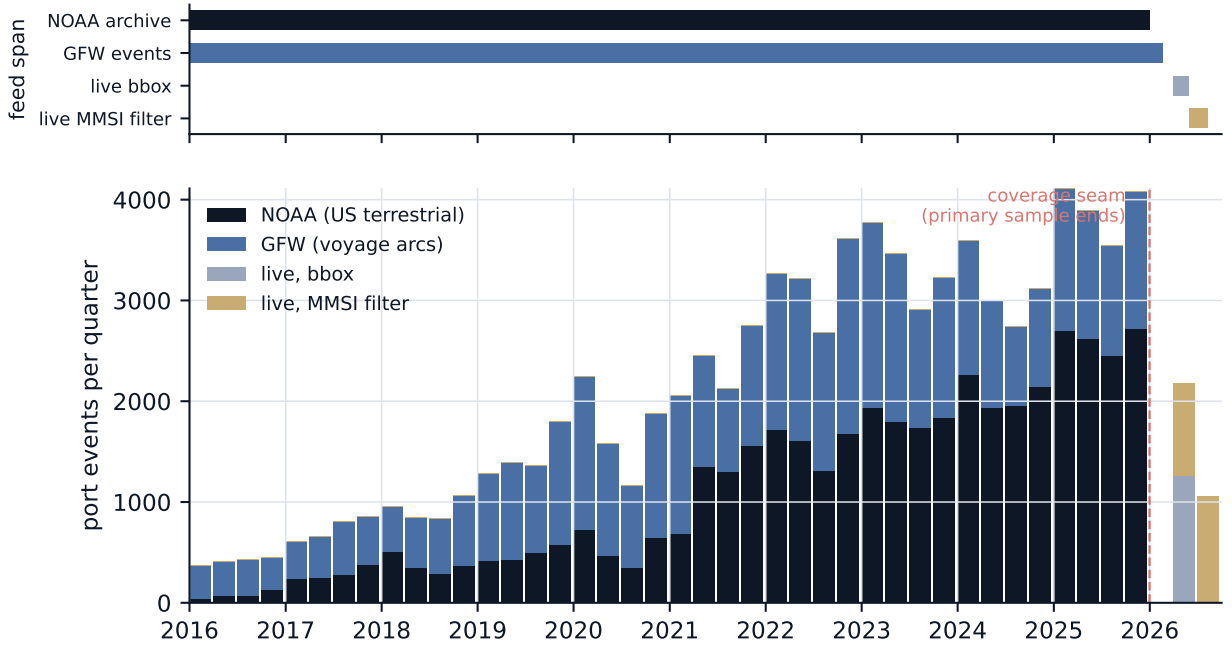


Figure 12: Port events per quarter by the feed that produced them, with the span of each feed above. The NOAA and GFW backfills carry the decade; the live feed, split at the cutover from a bounding-box subscription to a vessel-identifier filter, carries only the tail. The dashed line is the coverage seam at which the primary sample ends.

Table 3: Pre-registration entries. The commit column is discussed in the text.

entry	dated	what was fixed before the fit	commit
D-001–D-003	2026-08-10	A1 target, kernel, replay protocol	4239d13
D-016	2026-08-12	A2 design lock	4239d13
D-019	2026-08-12	A5 design lock	4239d13
D-022	2026-08-12	A4 design lock	4239d13
D-025	2026-08-12	H1 horizon test pre-specified	4239d13
D-028	2026-09-05	Part B null + FWL: target, controls, HAC lag, signs	e483a1b
D-031	2026-09-05	H2/H3/H4: signs, Bonferroni bar, holdout	c489550

up and over-shooting on the way down. A2 is +84.0% worse because it over-extrapolates the trend: the log link on an expanding window places current features several standard deviations above the training mean and the exponential converts that distance into multiplicative overshoot. A2 also falsified one of its pre-registered signs: ballast arrivals entered with coefficient -0.049 against a prior of positive. A4, the Kalman local-level model, is +0.7% from the null, which we treat as a tie. The tie carries information. A4 fitted its smoothing constant by maximum likelihood and chose $\alpha = 0.251$, an effective window of 7.0 weeks. For a series that is a local level plus noise, the exponentially weighted moving average is the optimal linear predictor [8]. The fitted model landing next to a four-week simple average indicates that the optimum was found, and not that four methods happened to tie. On this target and at this horizon, the naive mean is close to optimal, and

a fifth method would not have changed the conclusion. The sequence is not a monotone convergence: A2 was the worst of the four, not the middle rung.

A5: outage detection. A5 tested Bayesian online change-point detection [1] on each US terminal’s daily laden-departure count against 16 labelled outages over 54 terminal-years. The observation model is Poisson with a Gamma prior on the rate, and the detector maintains a posterior over the run length r_t (days since the last change) by

$$\begin{aligned} P(r_t = r + 1) &\propto P(r_{t-1} = r) \pi(y_t | r) (1 - H), \\ P(r_t = 0) &\propto \sum_r P(r_{t-1} = r) \pi(y_t | r) H, \end{aligned} \quad (10)$$

where $\pi(y_t | r)$ is the run’s negative-binomial posterior predictive and H the hazard; it alarms when the run-length-averaged rate falls below a fixed fraction of the

Table 4: One-week-ahead nowcasts of weekly US loadings (A2, A4) and conditional EU arrivals (A1), scored walk-forward from 2018 on the NOAA feed. MAE is in cargoes per week; the excess column is Equation (6). All three fail the pre-registered bar of beating both nulls.

model	MAE	persistence	climatology	vs climatology	80% cov.
A1 arrival-count climatology	2.314	2.217	1.814	+27.5%	73.7%
A2 negative-binomial GLM	3.961	2.551	2.152	+84.0%	75.7%
A4 Kalman local level	2.104	2.694	2.088	+0.7%	72.6%

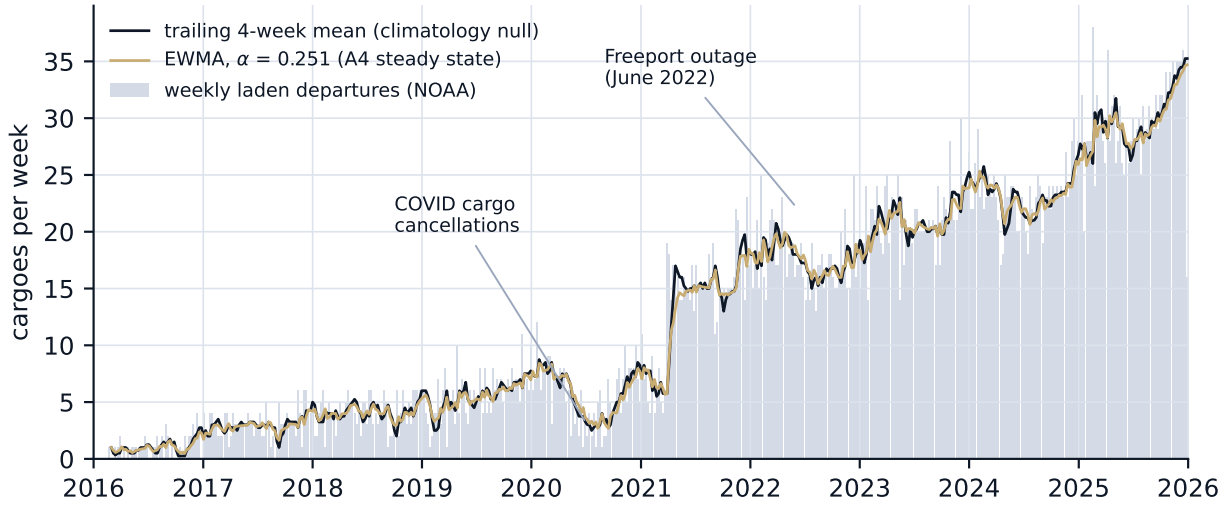


Figure 13: Weekly laden departures from US export terminals on the NOAA feed, with the two objects the nowcasts compete against: the trailing four-week mean (the climatology null) and the exponentially weighted moving average at the smoothing constant A4 fitted by maximum likelihood, $\alpha = 0.251$. The series peaks at 38 cargoes in a week and averages 29.8 per week in 2025. The 2020 trough is the COVID cargo-cancellation wave and the mid-2022 step down is the Freeport outage.

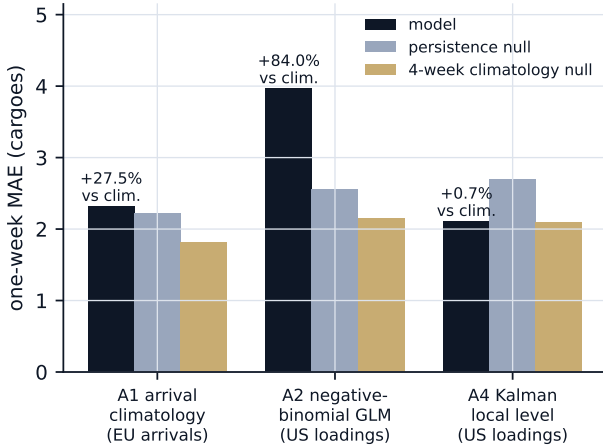


Figure 14: One-week MAE of each model against its two nulls, from Table 4. No model beats the four-week climatology; A4 ties it.

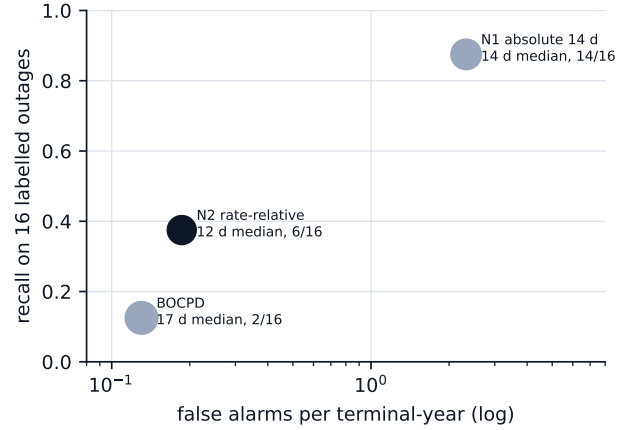


Figure 15: Recall against false alarms for the three detectors; marker size grows with median delay. N1’s recall is near-tautological, since a labelled outage is by construction a gap of at least fourteen days, and its false-alarm rate is the real story.

pre-change rate. The nulls are two silence rules that read the time since the last departure: N1 alarms after a fixed 14 days, N2 after five times the terminal’s own baseline inter-departure interval. Table 5 and Figure 15 give the scorecard. BOCPD did not outperform the rate-relative rule, because daily binning discards the exact timing that a point process carries: about half of normal days are zero-count at a two-day cadence, so each zero is weak evidence. N2 detects with a median delay of 12 days at 0.19 false alarms per terminal-year, with recall 38% (6/16); every miss is a slow-baseline terminal whose five-times rule exceeds the 21-day detection window. It is a usable but limited monitor, and it is the baseline rather than the proposed model. Its use should be stated plainly: a twelve-day median delay is slower than public news of a terminal outage, so N2 is an automated cross-check on the reconstruction rather than an alert a trader would act on.

A withdrawn test. A natural extension asks whether skill rises with horizon, since cargo at sea today determines arrivals two to three weeks ahead. This was pre-specified before any longer-horizon data was examined, with the premise, also written down, that the nulls would degrade with horizon. The test extends the A1 window index to $k = 1, \dots, 4$ and scores each W_k against the same two nulls. Extending to four weeks produced a +86.7% improvement over the best null at week four (Table 6). The improvement is an artefact, and Figure 16 shows why. A1’s target is conditional on legs already at sea at t , whereas both nulls are the last elapsed *unconditional* weekly count. The two coincide at one and two weeks, because no US-to-EU voyage completes in under seven days, so nothing arrives in the first week that had not already departed. Past the voyage tail the conditional truth falls toward zero (mean 5.89 arrivals in week one and 0.75 in week four) while the nulls continue to predict the full weekly rate

Table 5: Outage detection against labelled terminal outages. N2 is the pre-registered null and the usable artefact; its recall is the relevant number.

detector	recall	median delay (d)	false alarms / terminal-yr
BOCPD (pre-registered)	12% (2/16)	17	0.13
N1 absolute silence (14 d)	88% (14/16)	14	2.33
N2 rate-relative silence	38% (6/16)	12	0.19

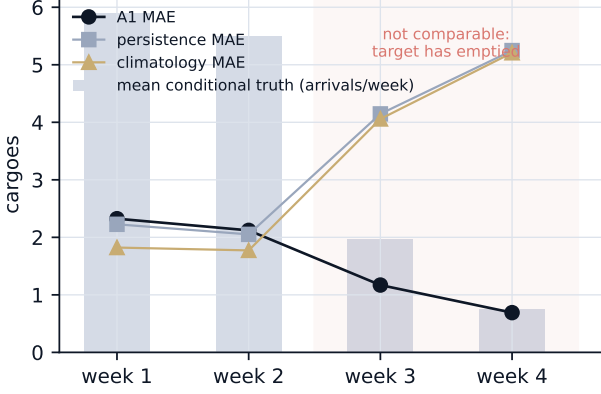


Figure 16: The withdrawn horizon test. Bars are the mean conditional truth in each window; lines are the MAE of A1 and of the two nulls. Beyond week two the target empties while the nulls keep forecasting a full week’s arrivals, so the apparent skill measures the target changing meaning.

(5.25 MAE). A1 was forecasting a smaller quantity, not forecasting better; a forecast of zero would have scored almost as well. The pre-specified premise fails, and the protocol treats that outcome as grounds to abandon the test. In the comparable region, the excess moves from -27.5% to -19.6% , which is the predicted direction but remains below the null. A valid horizon test needs an unconditional target, which requires composing A1 with a departure-process model such as A2; that composition is listed as future work.

6 The spread

6.1 Target, grid and controls

The target is the spread in \$/MMBtu. Henry Hub is quoted in \$/MMBtu and TTF in EUR/MWh, so TTF is converted at the day’s exchange rate and 3.412 MWh per MMBtu before the difference is taken,

$$s_t = \text{HH}_t - \frac{\text{TTF}_t^{\text{EUR/MWh}}}{3.412} \times (\text{EUR/USD})_t. \quad (11)$$

Taking the difference before converting would leak the exchange rate into the target as a spurious signal. The spread is sampled weekly on Mondays from 2018, the same grid as the physical nowcasts. The scored quantity is the h -week-ahead change,

$$\Delta_h s_t = s_{t+7h} - s_t, \quad h \in \{1, 4\}, \quad (12)$$

where t is in days. The level is too persistent to score usefully, because a model that predicts the current value performs well on it and learns nothing. The primary sample runs to 2025-12-31 (Section 3.5) and contains 379 complete weeks.

The control block z_t consists of the spread’s non-tanker drivers: heating and cooling degree days for the US and north-west Europe, US and EU storage, Brent, and a winter dummy. The price legs that construct the target are excluded as regressors. Fits are walk-forward with an expanding window and a 104-week minimum. A training row is admissible only if its target closed strictly before the test week, with a further one-week embargo [6]; the purge removes the rows whose target window straddles the test date and would otherwise carry the future into the fit, and the embargo removes one further row as a margin. Whether the embargo costs anything was not measured separately. Its effect on a model with at least 104 training rows is one row in a hundred, and it protects the null and the signals identically, so it cannot manufacture or hide a difference between them.

Newey–West standard errors. On a weekly grid with $h = 4$, consecutive targets share three weeks, so the residuals are serially correlated by construction and an ordinary least-squares standard error is roughly $\sqrt{h}$ too small. Every standard error in this section is heteroskedasticity- and autocorrelation-consistent (HAC) in the form of Newey and West [9]: with residual-weighted regressor autocovariances $\hat{\Gamma}_j$ and lag $L = h + 1$,

$$\hat{V} = \hat{\Gamma}_0 + \sum_{j=1}^L \left(1 - \frac{j}{L+1}\right) (\hat{\Gamma}_j + \hat{\Gamma}'_j), \quad (13)$$

and a reported t -statistic is $\hat{\beta}$ divided by the square root of the corresponding element of $\hat{V}$. The lag was fixed in advance at $h + 1$ weeks, one more than the overlap, and is never chosen from the data.

6.2 The controls-only null is worse than no change

Three models form a ladder in which each is the null for the next:

$$\text{M0: } \hat{s}_{t+7h} = s_t, \quad (14)$$

$$\text{M1: } s_{t+7h} = a + \rho s_t + \varepsilon_t, \quad (15)$$

$$\text{M2: } s_{t+7h} = a + \rho s_t + \gamma' z_t + \varepsilon_t. \quad (16)$$

M0 predicts no change and is the random-walk null. M1 is an autoregression of order one in the level, which nests M0 at $\rho = 1$. M2 adds the controls and is the AR(1)+controls null that the tanker signals were pre-registered to beat. The change in Equation (12) is recovered from each model’s level prediction. *Skill* is the fractional improvement in MAE over the model’s null,

$$\text{skill}(m \mid \text{null}) = 1 - \frac{\text{MAE}_m}{\text{MAE}_{\text{null}}}, \quad (17)$$

Table 6: A1 extended to four weekly windows. The skill column at W3 and W4 is an artefact of a conditional target scored against unconditional nulls; the harness no longer reports it as a result.

window	mean truth	A1 MAE	persistence	climatology	“skill”	80% cov.
W1	5.89	2.324	2.226	1.822	−27.5%	73.5%
W2	5.49	2.121	2.052	1.773	−19.6%	76.0%
W3	1.98	1.170	4.146	4.058	+71.2%	85.9%
W4	0.75	0.691	5.247	5.208	+86.7%	94.2%

so that a negative skill means the model is worse than predicting no change.

Table 7 shows that M2 does not beat M0. Relative to a no-change forecast, M2 scores −9.3% at one week and −18.2% at four. It is worse at the longer horizon and, as Figure 17 shows, worse in each of 2023, 2024 and 2025 taken individually, so the result is not a 2022 artefact. This is the behaviour expected of an efficient and persistent price series: the level already incorporates the fundamentals, so adding them as regressors introduces estimation noise. The operative null for the signals is therefore M0, a stricter bar than the one pre-registered.

6.3 No signal survives partialling

The test for each of the eight pre-registered signals T is the Frisch–Waugh–Lovell partial effect [3, 7]. With Z the matrix of M2’s regressors (the lagged level and the controls) and $M_Z = I - Z(Z'Z)^{-1}Z'$ the residual-maker, both the target and the signal are residualised on Z and the residuals regressed on each other,

$$\tilde{y} = M_Z y, \quad \tilde{T} = M_Z T, \quad \hat{\beta} = \frac{\tilde{T}'\tilde{y}}{\tilde{T}'\tilde{T}}, \quad (18)$$

with the t -statistic from Equation (13). By the theorem, $\hat{\beta}$ equals the signal’s coefficient in the full regression of y on $[Z, T]$, and that equality serves as the harness’s correctness test. The partial R^2 is the R^2 of the residual-on-residual regression: the share of the variance the controls leave unexplained that the signal explains. Partialling is not optional here. Both the signals and the spread carry a winter cycle (gas in transit runs about $1.35\times$ higher in winter, Figure 18), for unrelated reasons, and a regression of one on the other without the degree-day controls rediscovers winter.

Each signal carried a sign fixed in advance (Appendix A). A signal counted as carrying information only if all three conditions held:

1. $|t| > 1.96$ on the Newey–West standard error, at either horizon;
2. the sign of $\hat{\beta}$ matched the pre-registered sign;
3. the same sign held in at least two-thirds of individual calendar years, each year’s partial effect being estimated on that year’s weeks alone.

The last condition guards against 2022 alone driving a pooled fit.

No signal meets the bar (Figure 19, Table 8, Figure 20). The largest $|t|$ at either horizon is 1.50, and the largest partial R^2 is 0.009. The most direct mechanical hypothesis, that cargo at sea today lands in Europe in two to three weeks, was tested as the EU-bound stock and gave $t = +0.18$ at one week and $+1.40$

at four. At four weeks the four flow signals all have their pre-registered sign, but all lie well inside the noise band. They are directionally consistent with the mechanism and uniformly insignificant. No sign was falsified, because no estimate reached significance.

6.4 Three mechanisms a linear scan could miss

A one-at-a-time linear scan has three identifiable blind spots. Each was tested with a derived feature or a sample split, so each costs a single test. The three were pre-registered together with a Bonferroni bar, a primary horizon of four weeks chosen from the voyage mechanism, and a holdout (2025-01-06 to 2025-12-01, 48 weeks) that was not examined until a hypothesis had passed on the discovery sample (2018-01-01 to 2024-12-30, 327 weeks). The Bonferroni bar divides the significance level by the number of primary tests, $\alpha = 0.05/3$, so that the chance of at least one false positive across the three stays at 5%; the corresponding two-sided threshold is $|t| > 2.394$. A pass required the holdout sign to match.

H2 tests the share of cargo bound for Europe in place of the level,

$$\text{eu_share}_t = \frac{T_t^{\text{eu}}}{T_t^{\text{eu}} + T_t^{\text{unk}}}, \quad (19)$$

where T^{unk} is the unresolved-destination band. Most of the at-sea stock has no resolved destination, so two collinear levels with offsetting effects could each appear insignificant while their ratio carries the signal. A pre-registered gate voids the test if the share is mostly a decade-long trend, which would indicate improving coverage rather than changing cargo routing. The trend explains $R^2 = 0.209$ and the gate passes. H3 tests an interaction with EU storage: the regressors are T_t^{eu} , storage g_t (centred) and their product, with the partial effect taken on the product and both main effects moved into the control block. Additional cargo arriving when storage is 90% full has little effect, while the same cargo at 30% is highly price-relevant, and a linear fit averages the two states; the pre-registered sign on the product is negative. H4 tests a tightness split, the simplest form of a regime model. The scan is re-run within weeks in which EU storage is below its day-of-year seasonal norm, with the norm fitted on the discovery sample only, and the prediction is that the effect is positive in those weeks and larger than in the remaining weeks.

All three fail (Table 9, Figure 21), and none approaches the bar. The largest $|t|$ among the primary tests is 0.89. Each failure argues against a further attempt. The H2 estimate is stable and negative on both

Table 7: The model ladder on the primary sample. Skill is Equation (17) against M0; a negative value means worse than a no-change forecast.

h (wk)	model	n	MAE	RMSE	80% cov.	skill vs M0
1	M0	272	2.034	4.465	81.6%	–
1	M1	272	2.061	4.571	81.6%	–1.4%
1	M2	272	2.223	4.630	82.4%	–9.3%
4	M0	266	3.942	7.203	72.9%	–
4	M1	266	4.196	7.585	70.7%	–6.4%
4	M2	266	4.660	7.437	69.5%	–18.2%

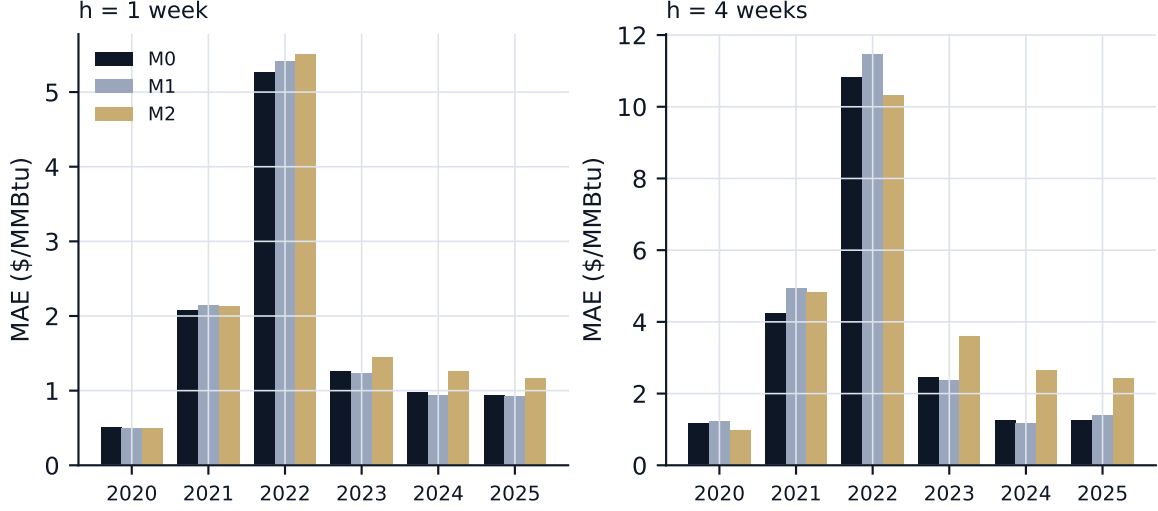


Figure 17: Per-year MAE of the ladder on the primary sample at both horizons. 2022 dominates the pooled numbers, which is why the year-consistency condition below exists; the controls-only M2 is worse than no-change in every calm year after it.

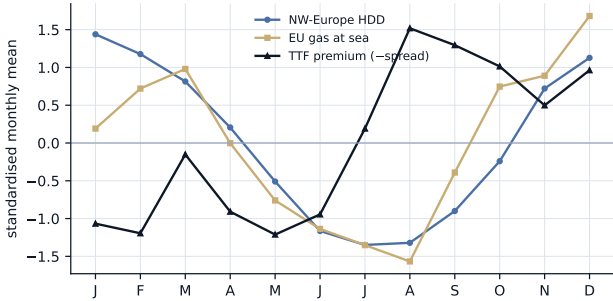


Figure 18: Why partialling is necessary: the tanker signal, the weather and the TTF premium share a season. Standardised monthly means over the panel.

discovery (-2.64×10^0) and holdout (-2.51×10^0), which is the wrong sign, and it never approaches significance. H3 is positive ($t = +0.46$) against a prior of negative, so neither the magnitude nor the direction of the storage interaction is present. H4's tight-regime effect is negative ($t = -0.17$) and smaller in magnitude than the loose-regime effect, which reverses both predictions. On this evidence, cargo arrivals do not have a measurable effect in either storage state.

Why they fail. The reasons are not a defect in the data. The pipeline is validated against the EIA total, the physical nowcasts show the signals track the quantities they measure, and every failure above is a stable near-zero rather than a sign that wanders. Three things explain a true null of this size. The scale of

the quantity is small: Section 7 shows the EU-bound stock at sea is about two days of European demand, so a one-standard-deviation change in it is a day and a half of supply against a price that moves on weather revisions. The information is not private: the marginal trader already has satellite-augmented positions and destination inference, so public terrestrial data arrives with a delay that the market has already discounted. And the measurement is attenuated: three-quarters of the at-sea stock has no resolved destination, and errors in a regressor bias its coefficient toward zero. The first two make the null the expected answer; the third means a small real effect could be hiding under the power floor, which is exactly what Section 7 says the design cannot exclude.

The linear null of the previous subsection has thus been tested against the three specific ways in which it could have been wrong, using pre-registered signs, a corrected bar and an untouched holdout, and it stands.

7 What the design could have found

A null result is informative only in proportion to the smallest effect it could have detected. For a partial effect estimated on n observations, $t^2 \approx n R^2 / (1 - R^2)$,

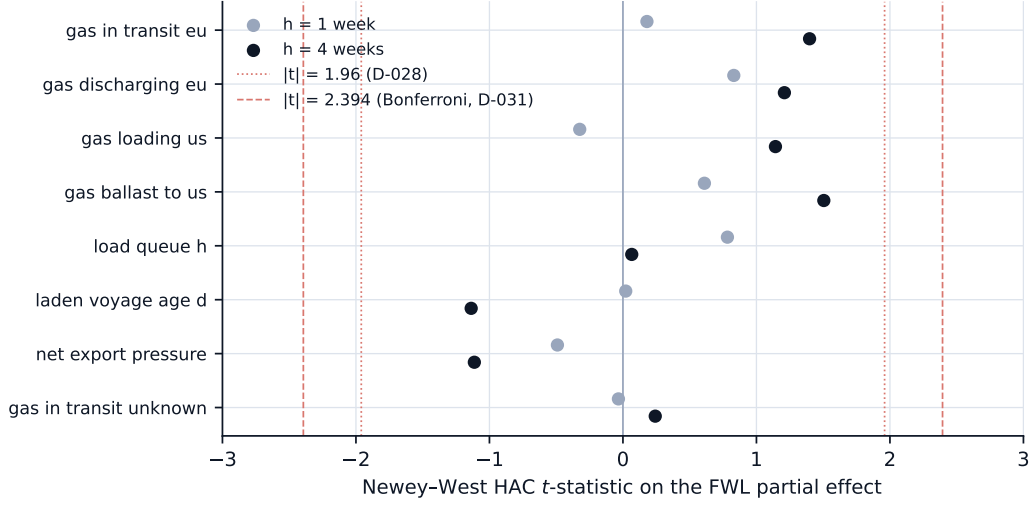


Figure 19: Newey–West t -statistic on the partial effect of each pre-registered signal, Equation (18), after controlling for persistence, weather, storage and oil. Light markers are the one-week horizon and dark markers the four-week horizon. The dotted lines at ± 1.96 are condition 1 above; the dashed lines at ± 2.394 are the Bonferroni bar of Section 6.4. A signal would need to lie outside the dotted lines, on the side of its prior, to count.

Table 8: FWL partial effects on the primary sample. Year consistency is the share of individual years whose partial effect has the full-sample sign.

signal	prior	$h = 1$		$h = 4$		year cons.
		HAC t	partial R^2	HAC t	partial R^2	
gas in transit, EU-bound	+	+0.18	0.000	+1.40	0.009	62%
gas in transit, unresolved	none	-0.03	0.000	+0.24	0.000	50%
US loading rate	+	-0.32	0.000	+1.14	0.006	62%
EU discharge rate	+	+0.83	0.001	+1.21	0.003	62%
ballast returning to US	+	+0.61	0.001	+1.50	0.008	62%
laden voyage age	−	+0.02	0.000	-1.14	0.004	62%
US load-queue wait	−	+0.78	0.001	+0.07	0.000	38%
net export pressure	+	-0.49	0.000	-1.11	0.003	75%

so at a fixed bar t^* the smallest detectable partial R^2 is

$$R_{\min}^2 = \frac{t^{*2}}{n + t^{*2}}, \quad (20)$$

which falls as $1/n$. With 327 discovery weeks and the Bonferroni bar of $|t| > 2.394$, the floor is 1.72% of residual variance (Figure 22). The largest partial R^2 observed anywhere in the paper, across eight signals, two horizons, three mechanisms and two samples, is 0.009.

The scope of the result follows from this. It excludes effects large enough to be traded and is silent on smaller ones. A signal explaining a third of a percent of residual variance would be physically real and commercially useless, and this design could not distinguish it from zero. Two facts about scale make the null less surprising than the intuition that motivated the study. The EU-bound stock of gas at sea averages 3.69 million m^3 of LNG, which is about 2.21 bcm of gas or 2.3 days of EU

demand at roughly 350 bcm per year. A one-standard-deviation change in that stock is 1.37 days of demand. The weekly spread has a standard deviation of \$11.22 per MMBtu. The question was therefore whether about a day and a half of marginal supply moves a price that responds to revisions in the weather forecast.

Distinguishing a physically real signal from a tradeable one. The distinction is worth keeping because it is where the interesting work now lies. A signal is physically real if it predicts the quantity it measures: Part A is that test, and the at-sea stock does predict European arrivals, with a lead equal to the voyage time, because that is arithmetic. A signal is tradeable if its effect on the price survives the controls at a size above the floor of Equation (20). The route to the first question is more physical targets scored against published totals; the route to the second is a lower floor, which means a

Table 9: The three pre-registered mechanism tests on the discovery sample at $h = 4$ with Bonferroni bar $|t| > 2.394$. None is significant. H3 has the wrong sign, and H4’s tight-regime effect is smaller than its loose-regime effect.

hypothesis	prior	$\hat{\beta}$ (discovery)	HAC t	year cons.	partial R^2	$\hat{\beta}$ (holdout)
H2: EU share of at-sea stock	+	-2.64×10^0	-0.89	71%	0.003	-2.51×10^0
H3: transit $\times$ EU storage	−	4.78×10^{-9}	+0.46	71%	0.001	-1.49×10^{-9}
H4: transit, tight regime	+	-1.13×10^{-7}	-0.17	100%	0.000	2.71×10^{-7}
H4: transit, loose regime	+	-2.25×10^{-7}	-1.00	100%	0.009	−

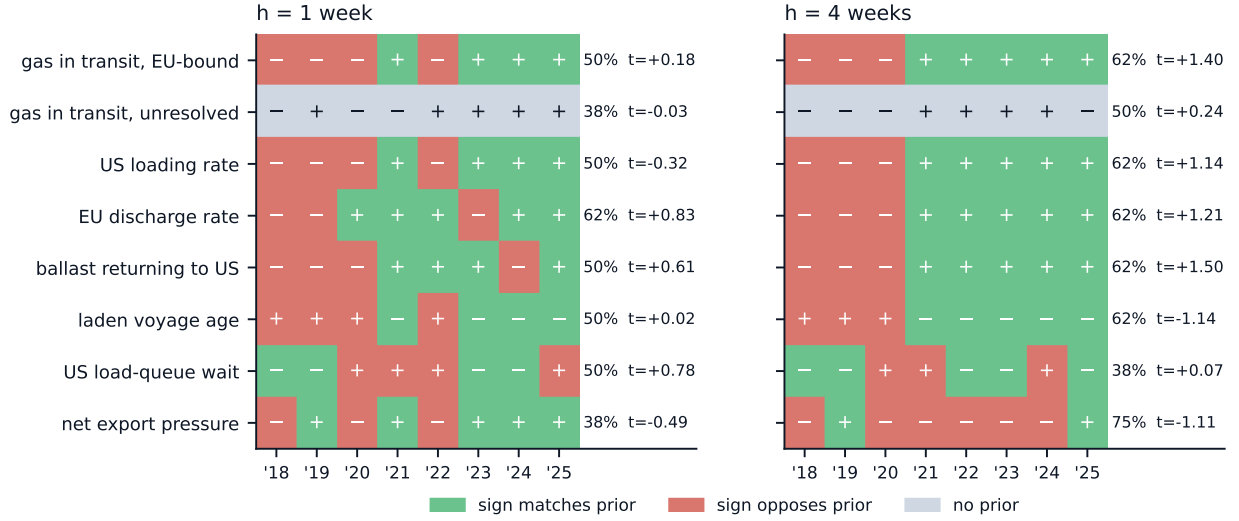


Figure 20: Condition 3 by signal and year. Each cell is the sign of the partial effect estimated on that year’s weeks alone, green where it matches the pre-registered prior and red where it opposes it; the unresolved band has no prior. The right-hand labels give the year consistency and the pooled t . At four weeks the flow signals carry their prior sign only from 2021 onward.

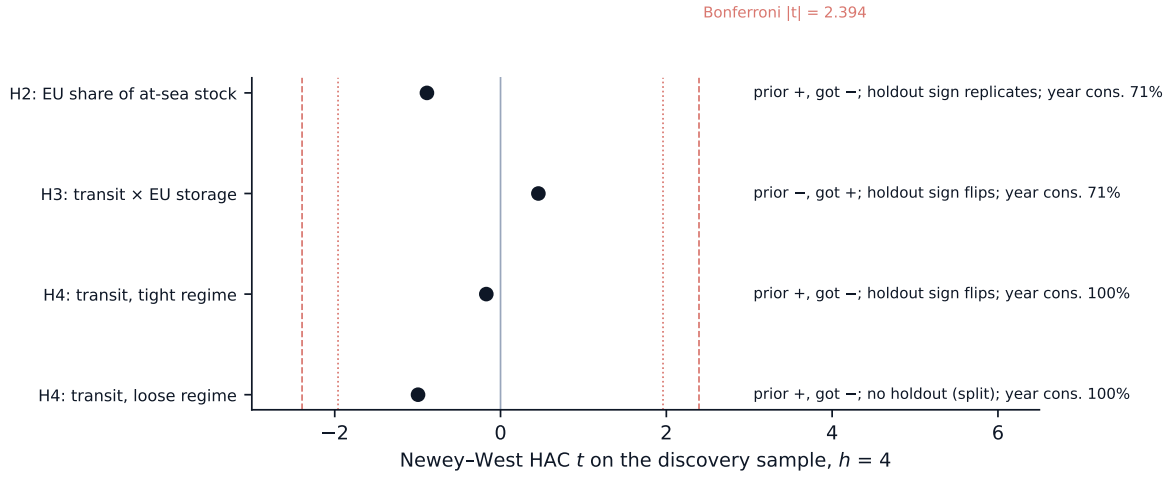


Figure 21: The mechanism tests against the Bonferroni bar (dashed) and the ordinary bar (dotted). The annotation gives the prior sign, the sign obtained, whether the holdout coefficient carried the same sign, and the year consistency.

longer coverage-consistent sample or a less attenuated regressor, not a different model.

8 Discussion

The null is the outcome an efficient market predicts, and it is important to be precise about which question was asked. LNG cargo tracking is a commercial industry. Several vendors sell satellite-augmented AIS with proprietary destination inference to gas trading desks in real time. The signals in this paper are built from public terrestrial AIS and a free voyage archive, which is a coarser version of what the marginal trader already has, and it reaches a public researcher with a delay that a desk would find unusable. The test was therefore not whether the physical signal is informative, but whether public physical data is mispriced in a liquid benchmark at one- to four-week horizons. A negative answer to that question is close to what one would expect in advance. What the paper adds is evidence that the negative result is not an artefact of a defective pipeline, an unlucky

specification, or a search that stopped at the first $t > 2$. The pipeline is validated against a published total, the specification was fixed before the fit, and the search was corrected for multiplicity and held out.

What the work found besides the null. A reader who takes only the headline would miss most of the content, so the other findings are listed here.

1. *The naive mean is near-optimal for weekly loadings, and the reason is known.* Four nowcasts approached a four-week moving average from above without crossing it, and the one with a fitted smoothing constant chose a seven-week window by maximum likelihood. For a slow industrial flow that is a local level plus noise, the exponentially weighted mean is the optimal linear predictor, so this follows from theory and should temper the expectation that a richer model will beat a moving average on any such series.
2. *A voyage-derived feature leaks by construction unless bounded.* The 65% of legs closed by the future at

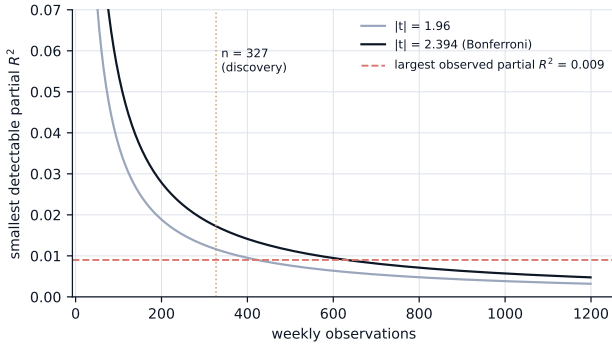


Figure 22: Smallest detectable partial R^2 , Equation (20), as a function of sample size at the two bars used in the paper, with the largest observed partial R^2 and the discovery sample size marked. Halving the floor needs twice the sample, which at a weekly grid means another six years of coverage-consistent data.

a 2020 as-of date was not a bug anyone wrote on purpose; it is what the natural implementation does. The vintage hazard is subtler still, since a date filter passes. Both are cheap to detect with a print log and, to our knowledge, rarely checked.

3. *Coverage has a gradient, and it runs opposite ways in the two public archives.* The NOAA capture rate rises from under half to above unity over the decade because receiver density grew; GFW's falls over the same years. Neither archive is a complete record on its own and the union is not their sum.
4. *A pre-registered sign was falsified.* Ballast arrivals entered the loadings model with a stable negative coefficient against a prior of positive. A ship must arrive empty before it leaves full, so the sign is a puzzle about timing rather than mechanism, and it is recorded rather than explained.
5. *Almost half of matured laden departures never pair with a European arrival.* They pair instead with the vessel's return to the Gulf, with a median duration of about a month. These are cargoes whose destination the public feeds never observed, and they are the reason the arrival model could not assume that a leg still open is still Europe-bound.
6. *A coverage seam was found by a plot, not a test.* The 91% fall in the at-sea stock at the turn of 2026 passed a validation sweep of seventy checks and a test suite of several hundred tests. It was visible in the first figure drawn for this paper.
7. *An outage monitor with a stated recall came out of the exercise as the null, not the model.* A rate-relative silence rule detects with a median delay of 12 days at 0.19 false alarms per terminal-year and 38% recall.

None of these is a price result. All of them are results about the data, and they are the proof of concept the pipeline delivers: a decade of port events that reconcile with a published total, built point-in-time and checked against its own live prints. The growth in US export capacity that dominates every volume panel of Figure 11 is real and large, but it is public information from project sanctions and EIA reports, not something AIS reveals, and the spread scan removes it deliberately through the

persistence term.

9 What would change the answer

Several changes could alter the answer, and they divide into changes to the question and changes within it.

A private information advantage. Satellite coverage of the open ocean or a destination model that resolves the three-quarters of the stock this pipeline cannot would change the question rather than the answer, and would make the study a test of the vendor's product. Within public data the most promising destination feature is geometric: a Gulf cargo bound for Europe must exit north-east through the Straits of Florida while Asia-bound cargoes route south, and the project's notes record that a north-east crossing is strong evidence of a European arrival about thirteen days ahead. The crossing is inside terrestrial range, so it is a coverage question, and the live system's slot allocation was changed to keep departing vessels subscribed through the Straits. That feature was not available on the decade archives and is not tested here.

A fused capture-corrected supply series. NOAA and GFW have opposite coverage gradients (Section 2.4). A deduplicated union of the two, matched on vessel, terminal and day, with the residual under-count carried as an exposure offset calibrated to the EIA total, would recover the early years as levels rather than only as changes. The acceptance test is that the fused series reconciles with EIA within a few percent in every year.

A valid horizon test. The withdrawn test of Section 5 needs an unconditional arrival target: A1's conditional term composed with a departure-process term for cargoes not yet loaded, scored against nulls of the same unconditional quantity. The composition is a genuine modelling task and was parked because its downstream consumer, the spread, showed no response to the input it would sharpen.

The untested variants. Within public data, the remaining untested approaches are joint rather than marginal fits (spike-and-slab or partial least squares over the collinear block), a fitted regime model in place of an observable split, distributed lags of the signals, and the twenty-six signal keys the panel does not use. Each has a weaker prior than the three mechanisms already tested, and each faces a larger multiplicity problem against a residual that has never exceeded 0.009 of variance. Any of them should be accompanied by its own pre-registration, its own multiplicity correction, and the untouched-holdout standard used here.

A longer coverage-consistent sample. The power floor falls as $1/n$. Halving it requires roughly six further years on the weekly grid, or a daily grid with a target that is not a near-random walk at that cadence. The live

feed is the only source that continues, and its coverage is a fraction of the archives', so the seam of Section 3.5 will not close on its own.

Process. Two cheap changes are recorded as open work: a coverage-regime guard on the modelling panel so that a level break at a feed boundary fails a check rather than a plot, and a third-party timestamp on every future pre-registration entry.

10 Limitations

Every limitation below is one that a commercial gas desk has already paid to remove, which is the immediate disadvantage any public-data study of this market starts from. They are listed so that the scope of the null is exact.

Terrestrial AIS misses vessels and drops them.

The archives and the live feed see a vessel only within range of a shore receiver. Vessels that load out of range leave no departure, vessels at sea are invisible for the whole crossing, and the live feed additionally loses any vessel that does not hold a subscription slot. The rescue backstop of Section 2.3 quantifies the live shortfall: 2,133 candidate lookups were declined by the budget in the period shown, and 2,072 were made. The measured publication lag of a departure is negligible on the archives and up to a day on the live feed.

The polygons are hand-drawn. The 142 berth, anchorage and approach polygons were digitised by hand and are provisional at several terminals. An envelope drawn too tight emits spurious exits during a channel transit; one drawn too loose merges neighbouring terminals. The state machine's stickiness and berth-override rules limit the damage, and the capture check bounds the aggregate effect on the US side, but no per-terminal audit of the geometry has been done.

The two archives have opposite capture gradients. NOAA-only capture of US export cargoes is 46% in 2020 and 74% in 2018 (Table 2), and GFW's capture runs the other way. Early-year levels of every US signal are biased low. The walk-forward designs operate on within-window changes and the FWL scan on residuals, but a level-dependent non-linearity in the early years would be misread.

No European anchorage history. The GFW feed carries port visits but no anchorage events, so European queue and absorption signals exist only from the 2026 live feed. Four of the five composite signals originally pre-registered as features for Section 6 turned out to have between 9 and 46 daily observations and were excluded by a documented amendment before any spread fit. The at-sea and US-side mechanism was tested on the decade; the EU-congestion mechanism could not be tested.

Destination resolution. Between 75% and 83% of the at-sea stock has no resolved destination. The EU-bound series is a minority observed slice whose definition depends on coverage. The unresolved band is carried as its own feature, and the ratio of the two was tested as H2. Attenuation from this measurement error biases coefficients toward zero and cannot be excluded as a contributing factor.

The coverage seam. The backfill feeds end on 2025-12-31 and 2026-02-21. The full-panel results in Appendix B include the thin live tail. In that sample 24% of the holdout weeks lie after the seam, and one full-panel finding, a change of sign in the H2 holdout coefficient, did not survive truncation and is not claimed. The primary sample is truncated at the seam.

Regime pooling. Signals are read from the pooled regime tag, which is the only tag spanning the decade. A small residual over-count where NOAA and GFW both observe the same US departure is documented in the project's signal catalogue. A robustness run on the NOAA tag alone for the US-side signals was not performed.

The pre-registration trail. As Section 4 states, the git history does not independently establish that the log entries preceded the fits they govern.

Sample size and a dominant regime. The scored spread sample is 379 weeks, heavily autocorrelated, and 2022 alone spans a range larger than the remainder of the decade combined. The year-consistency condition and the holdout are the defences against this; a longer sample without a coverage seam would be preferable to either.

The outage monitor is slow. The usable detector has a twelve-day median delay. A terminal outage of that length is public news well before the monitor fires, so its value is as an automated check on the reconstruction, not as information.

Vintage residual. 68 rows in the live tail changed after a next-day print (Section 3.3). They lie outside the modelling sample, but they show that the knowable bound is exact on event time and only approximately exact on observation time in a thin-coverage regime.

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A Signal definitions and pre-registered signs

Table 10 lists the eight tanker signals used in Section 6, the aggregation applied when collapsing per-terminal or per-zone bands to one daily series, and the sign fixed before any fit. The sign convention is as follows. The spread $HH - TTF$ is negative in nearly every period, so a narrowing of the spread means a rise toward zero, which is a positive change.

Volumes are in m^3 of LNG. Loading and discharge rates are amortised m^3 per day across berth hours, Equation (4), so that a closed visit integrates to one cargo. Ages and waits are in days and hours respectively. The project’s signal catalogue documents the remaining twenty-six keys, the two bases, and the confidence columns (dispersion, open fraction and estimated fraction) that accompany every value.

A recorded inconsistency. The catalogue’s own text assigned net export pressure two incompatible signs in two places. The positive sign was adopted on the economic reasoning above and recorded as the prior. The data returned a negative point estimate at both horizons with $|t| \leq 1.6$, which neither confirms nor falsifies it.

B Full-panel results

The primary sample is truncated at the backfill horizon (2025-12-31). The same scans run to the data maximum (395 weeks) are reported here for completeness. They were the runs first recorded in the decision log, and they change no conclusion.

C Glossary of terms

AIS. The Automatic Identification System: a VHF broadcast every vessel over 300 gross tonnes must transmit, carrying its identity (MMSI), position, speed and course, and less often its draught, destination and ETA. Terrestrial receivers hear it within roughly 40 nautical miles of shore; satellites hear it at sea.

MMSI. Maritime Mobile Service Identity, the nine-digit number that keys AIS messages to a vessel. It can change on reflagging, which is why the fleet list is keyed on the IMO hull number and resolved to an MMSI.

FSRU. Floating storage and regasification unit, a converted carrier moored permanently as an import terminal.

Laden, ballast. A carrier is laden when carrying cargo and in ballast when empty; Equation (1) infers the state from draught.

Leg, visit, queue. The three event pairings: departure to next arrival (a voyage), mooring to departure (berth occupancy), first anchorage entry to mooring (the wait to berth).

Physical, knowable. The two bases every signal is built on: with hindsight, and using only what was observable on the day (Section 3).

Vintage. The version of a dataset as it existed on a given date. A feature has a vintage hazard when the table it comes from did not exist at the as-of date even though its rows carry earlier timestamps.

Persistence null, climatology null. Forecasting next week’s count as last week’s count, and as the trailing four-week mean, respectively.

MAE, skill, excess. Mean absolute error; the fractional MAE improvement over a null, Equation (17); and the fractional MAE deterioration relative to a null, Equation (6).

80% coverage. The share of realised outcomes that fell inside the model’s central 80% predictive interval. A calibrated model scores about 80%.

Walk-forward. Refitting the model at every step on data strictly before that step, so that every forecast is out of sample. Contrast with cross-validation, which shuffles time.

Table 10: The eight pre-registered signals, their aggregation and their prior sign.

signal	aggregation	mechanism	prior
gas in transit, EU-bound	sum over EU zones	more cargo landing in Europe soon; TTF falls	+
gas in transit, unresolved	sum	destination unknown; no directional prior	none
US loading rate	sum over US berths	gas leaves the US balance (HH firms) and reaches Europe (TTF falls)	+
EU discharge rate	sum over EU berths	Europe absorbing gas now; TTF falls	+
ballast returning to US	sum	tonnage returning to load; future exports rise	+
laden voyage age	n -weighted mean	cargo floating instead of landing; Europe stays tight	−
US load-queue wait	n -weighted mean	US cannot export; HH softens and Europe is short	−
net export pressure	n -weighted mean	$z(\text{loadings}) - z(\text{load queue})$; both components act in the same direction	+

Table 11: Model ladder, full panel.

h (wk)	model	n	MAE	RMSE	80% cov.	skill vs M0
1	M0	288	1.997	4.371	82.3%	−
1	M1	288	2.023	4.473	82.3%	−1.3%
1	M2	288	2.205	4.542	83.0%	−10.4%
4	M0	282	3.947	7.090	73.8%	−
4	M1	282	4.168	7.444	72.0%	−5.6%
4	M2	282	4.705	7.378	69.1%	−19.2%

Purge and embargo. Dropping training rows whose target window overlaps the test date, and one further row as a margin (Section 6).

HAC, Newey–West. A heteroskedasticity- and autocorrelation-consistent standard error, Equation (13), which corrects for the serial correlation that overlapping targets induce.

Bonferroni. Dividing the significance level by the number of tests so that the family-wise false-positive rate stays at the nominal level.

FWL, partial effect, partial R^2 . The Frisch–Waugh–Lovell construction of a coefficient net of controls, Equation (18), and the share of the residual variance that the signal explains.

Discovery, holdout. The sample on which a hypothesis is graded, and the final stretch left untouched until it has passed.

EWMA. Exponentially weighted moving average, $\hat{x}_t = \alpha y_t + (1 - \alpha)\hat{x}_{t-1}$.

BOCPD. Bayesian online change-point detection, Equation (10).

CRPS. Continuous ranked probability score, a proper score for a full predictive distribution that reduces to absolute error for a point forecast.

Capture rate. Observed cargoes divided by the EIA-implied count, Equation (2).

Coverage seam. The date after which only the thin live feed carries the signals (Section 3.5).

D Reproducibility, code, and the use of AI

Reproducibility. Every number in this paper is generated. The replays write machine-readable results (`make paper-results`). A generator converts them into the LaTeX tables and into a file of macros from which every in-prose number is drawn (`make paper-tables`). The figures are drawn from the same tables the models read (`make paper-figures`); the engineering figures and the macros for every engineering count in the prose are produced by a second generator (`python -m paper.figures2`) from the live tables. A number that is not generated cannot appear in the text, because the corresponding macro is undefined and the document does not compile. The underlying replays are `make a1-replay, a2-replay, a4-replay, a5-replay,`

Table 12: FWL partial effects, full panel. The largest $|t|$ here (1.72) belongs to the unresolved-destination band and falls to +0.24 on the primary sample. It was produced by the contaminated tail.

signal	prior	$h = 1$		$h = 4$		year cons.
		HAC t	partial R^2	HAC t	partial R^2	
gas in transit, EU-bound	+	+0.08	0.000	+1.03	0.004	75%
gas in transit, unresolved	none	-0.74	0.001	-1.72	0.009	62%
US loading rate	+	-0.19	0.000	+0.63	0.001	62%
EU discharge rate	+	+0.84	0.001	+0.46	0.000	62%
ballast returning to US	+	+0.14	0.000	+0.31	0.000	62%
laden voyage age	−	-0.04	0.000	-1.26	0.005	62%
US load-queue wait	−	+1.01	0.001	+0.55	0.001	38%
net export pressure	+	-0.82	0.001	-1.60	0.009	75%

Table 13: Mechanism tests, full panel. The holdout coefficient for H2 changes sign relative to the primary sample. In this sample 24% of the holdout weeks lie after the coverage seam, and the change of sign is not claimed as a finding.

hypothesis	prior	$\hat{\beta}$ (discovery)	HAC t	year cons.	partial R^2	$\hat{\beta}$ (holdout)
H2: EU share of at-sea stock	+	-2.64×10^0	-0.89	71%	0.003	8.23×10^0
H3: transit $\times$ EU storage	-	4.78×10^{-9}	+0.46	71%	0.001	6.73×10^{-10}
H4: transit, tight regime	+	-1.13×10^{-7}	-0.17	100%	0.000	9.21×10^{-7}
H4: transit, loose regime	+	-2.25×10^{-7}	-1.00	100%	0.009	-

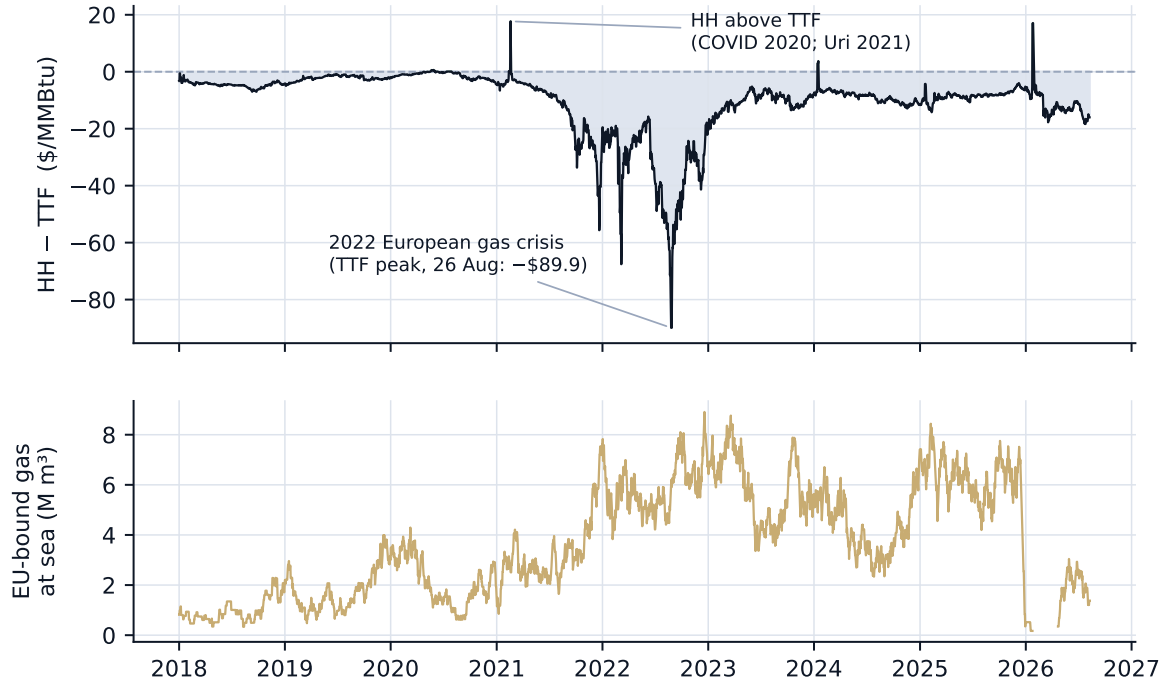


Figure 23: Figure 7 extended to the data maximum. The 91% step in the at-sea stock at the start of 2026 is the coverage seam and not a physical event.

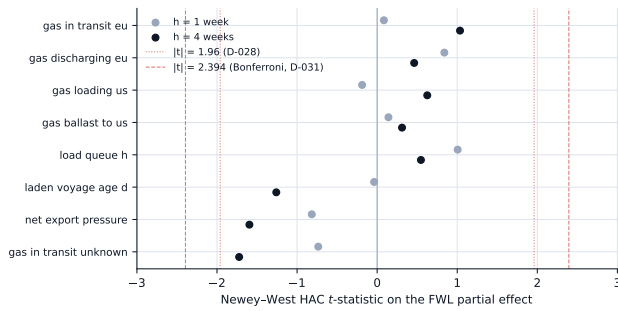


Figure 24: Figure 19 on the full panel.

a1-h1, b0-coverage and mechanisms-coverage, together with their full-panel variants. The panel itself is produced by `make signals` followed by `make model-panel`. The source, the decision log and the test suite (591 tests at the time of writing) are available in the project repository.

Data and licences. AIS positions are from NOAA’s Marine Cadastre and Global Fishing Watch under their respective terms. EIA, FRED, GIE AGSI+ and Open-Meteo are free public APIs. The World Bank monthly series is licensed CC BY 4.0 and is the only price series redistributed with the code. The daily TTF series is obtained from a public quote feed under personal-use terms and is not redistributed; the loader retrieves it.

Use of AI. The author used an AI coding assistant (Anthropic’s Claude, through Claude Code) extensively in this project. A substantial fraction of the pipeline, the modelling harnesses, the tests and the first drafts of this text were written with its assistance. Every design decision, hypothesis, sign and acceptance bar in the decision log is the author’s. Every result was reproduced by the author from the stated `make` targets, and the author takes full responsibility for the content. The assistant was also used to audit the project’s own claims before this paper was written, which produced the corrections recorded in the final entries of the decision log.